

BAYESIAN PRODUCTION MODELS APPLIED TO NORTH ATLANTIC MAKO SHARKS USING JABBA AND JABBA-SELECT, UPDATED IN 2026

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SUMMARY

The surplus production models used in the 2025 shortfin mako assessment in the North Atlantic were updated to include an additional year of catch and CPUE data, as well as revised life history parameters under the three growth scenarios agreed by the working group. The demographic priors were updated with revised natural mortalities. The base case included M values adjusted to reduce the number of animals surviving to the last time period. A sensitivity analysis used the M values from the 2025 SS3 runs. For both JABBA and JABBA-Select, the CPUE series implied a decrease-increase-decrease trend in biomass. However, the different scenarios implied different productivities, leading to different estimates of the potential yield and current overfishing status across scenarios.

RÉSUMÉ

Les modèles de production excédentaire utilisés dans l'évaluation de 2025 du requin-taureau bleu de l'Atlantique Nord ont été mis à jour afin d'intégrer une année supplémentaire de données de capture et de CPUE, ainsi que des paramètres du cycle vital révisés dans le cadre des trois scénarios de croissance retenus par le groupe de travail. Les distributions a priori démographiques ont été mises à jour à la lumière des taux de mortalité naturelle révisés. Le cas de base comprenait des valeurs M ajustées afin de réduire le nombre d'animaux survivant jusqu'à la dernière période. Une analyse de sensibilité a été réalisée à partir des valeurs M issues des scénarios de SS3 de 2025. Tant pour JABBA que pour JABBA-Select, la série de CPUE a révélé une tendance à la baisse, puis à la hausse, puis à nouveau à la baisse de la biomasse. Toutefois, les différents scénarios impliquaient des productivités différentes, ce qui a conduit à des estimations différentes de la production potentielle et de l'état actuel de la surpêche selon les scénarios.

RESUMEN

Los modelos de producción excedente utilizados en la evaluación del marrajo carite de 2025 en el Atlántico norte se actualizaron para incluir un año adicional de datos sobre capturas y CPUE, así como parámetros revisados del ciclo vital según los tres escenarios de crecimiento acordados por el Grupo. Las distribuciones previas demográficas se actualizaron con las tasas de mortalidad natural revisadas. El caso base incluía valores de M ajustados para reducir el número de animales que sobrevivían hasta el último periodo. En un análisis de sensibilidad se utilizaron los valores M de las simulaciones SS3 de 2025. Tanto en el caso de JABBA como de JABBA-Select, la serie de CPUE mostró una tendencia de disminución-aumento-disminución en la biomasa. Sin embargo, los distintos escenarios implicaban diferentes niveles de productividad, lo que daba lugar a estimaciones diferentes del rendimiento potencial y de la situación actual de sobrepesca en cada uno de ellos.

KEYWORDS

Mathematical models; stochastic models; stock assessment; population dynamics

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1. Introduction

At the 2025 north Atlantic mako stock assessment (ICCAT, 2025), the JABBA surplus production model was used as an alternative to the SS3 models (Winker *et al.*, 2018; Babcock, 2025). Many sensitivities were run, with different priors for carrying capacity (K), process error assumptions, and other settings, in addition to the three life history scenarios agreed by the data meeting (ICCAT, 2025). A JABBA-Select model was also run for all three scenarios with several alternative priors for the unfished spawning stock biomass (Winker *et al.*, 2020; Babcock, 2025). This update includes catch and CPUE for an additional year (2024), revised CPUE analyses, including a combined index developed intersessionally, and revisions of the three life history scenarios. As in 2025, these production models are not expected to be used for management advice but are intended to provide an alternative view on the population dynamics and recent trends in abundance. This paper has been revised to report the runs that were conducted at the June 2026 mako assessment, correcting an error that was identified in the JABBA software at the meeting and also including some new sensitivities.

2. Methods

2.1 JABBA

For the updated JABBA runs, the catch data were the same as in 2025 with one additional year. The CPUE data were used were the same as the 2026 SS3 models and were quite different from the 2025 indices (**Figure 1**). The Japan and Morocco series were unchanged, but most of the other indices were replaced by a combined index developed in 2026 (Coelho *et al.*, 2026). The Spanish longline index was only used through 2017 as Spain was included in the combined index after 2018. As in the reference runs from 2025, the settings for the JABBA models included observation error in catch (prior log-sd = 0.1), process error was fixed at log-sd = 0.05, and observation error for the CPUE was calculated as the standard error calculated from the input CVs for each point (increased to 0.2 if the estimate was below 0.2), plus an estimated parameter for each series with a gamma prior. The prior for K was lognormal with a mean of 200,000 and a CV of 2.0.

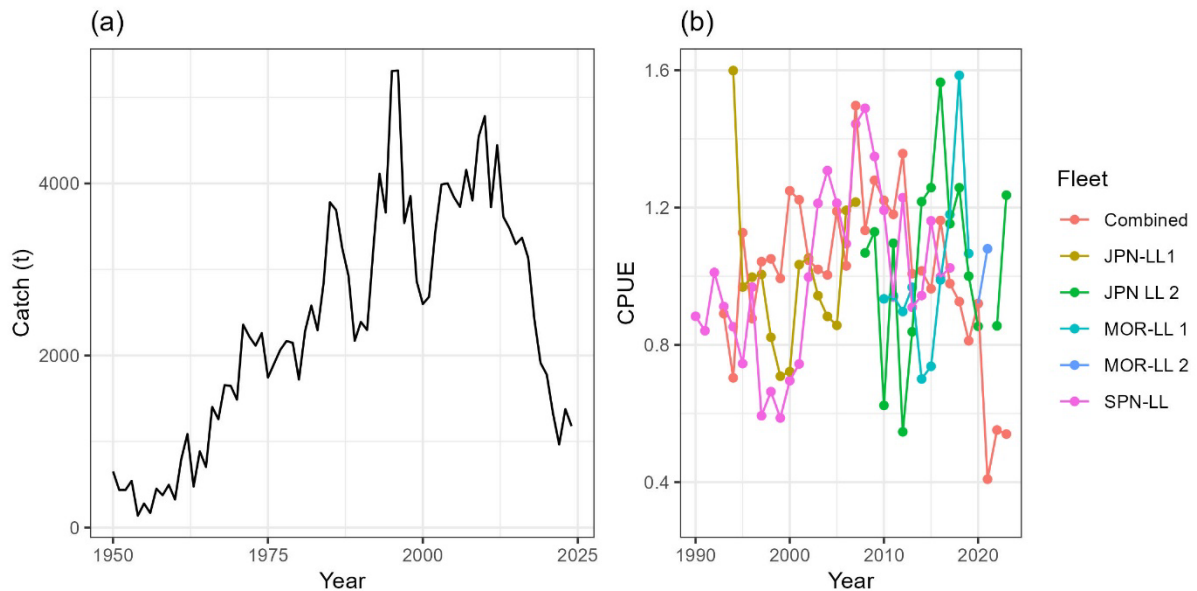


Figure 1. (a) Catch and (b) CPUE indices (divided by their means) for north Atlantic shortfin mako updated in 2026.

The three life history scenarios identified by the working group were all considered equally. The scenarios were (**Table 1**): (1) growth at two bands per year, (2) growth based on mark-recapture data; and (3) growth similar to 2017 assumptions. These scenarios influenced the biomass relative to K that maximizes yield in the Pella-Tomlinson surplus production model, which was fixed to the value calculated with the deterministic demographic analysis and the r prior, which had a value calculated from the deterministic

demographic analysis and CV of 0.2 to encompass uncertainty (Bowlby and Babcock, 2026). Each scenario had both a base case and a sensitivity analysis. The base case used priors derived from demographic analysis using the adjusted estimates of M used in the SS3 base cases, which were adjusted to reduce the total population to 0.4% in the oldest age group while the sensitivities used the original SS3 mortality values from the 2025 runs (Bowlby and Babcock, 2026). The JABBA software allows a maximum value of the Pella-Tomlinson m parameter of 10, which is a $Bmsy/K$ of 0.774. So, the model used a value of 0.774 for base scenario 2 and sensitivity scenario 3 (**Table 1**).

Table 1. Inputs for the JABBA and JABBA-Select scenarios revised in 2026 from Bowlby and Babcock (2026), except for M which was calculated from longevity ($M = -\log(0.004)/tmax$). Values above the maximum allowed in JABBA (0.774) are replaced with 0.774 in JABBA runs.

<i>Case</i>	<i>Scenario</i>	<i>r</i>	<i>Bmsy/K</i>	<i>steepness (h)</i>	<i>longevity (tmax)</i>	<i>M</i>
Base	1	0.051	0.670	0.317	42	0.131
	2	0.020	0.846	0.196	36	0.153
	3	0.019	0.745	0.253	69	0.080
Sensitivity	1	0.102	0.541	0.573	42	0.077
	2	0.094	0.557	0.504	36	0.089
	3	0.015	0.793	0.228	69	0.047

To evaluate the relative importance of the data versus the priors, a post-model pre-data run (i.e. catch-only model) was conducted for all six models, using only the catch data and no CPUE indices, as well as runs excluding each index one at a time. Retrospective analysis and hindcast cross-validation were done back 9 years from the current end year (2016 to 2025) and Mohn's rho and MASE were used to check for retrospective bias and prediction skill, respectively (Carvalho *et al.* 2021). Model convergence was checked with MCMC traceplots and the Geweke P diagnostic.

Additional sensitivities conducted at the 2026 assessment included: (1) using the same prior for r and value of $Bmsy/K$ as was used in 2025 (Scenario 1: $r = 0.085$, $Bmsy/K = 0.597$; Scenario 2: $r = 0.085$, $Bmsy/K = 0.59$; Scenario 3: $r = 0.044$, $Bmsy/K = 0.66$), (2) the original base case priors but with a break in the combined index between 2020 and 2021 in case the change in management influenced the CPUE series, (3) the prior that was used in one run in 2019 ($r = 0.0254$, $Bmsy/K = 0.5$); and (4) the prior that was used for the South Atlantic in 2025 ($r = 0.114$, $Bmsy/K = 0.578$).

2.2 JABBA-Select

JABBA-Select was run for the same three life history scenarios (**Table 1**, **Table 2**), updated by Bowlby and Babcock (2026), and the updated catch and CPUE data along with selectivities for each fleet taken from the current SS3 assessment, scenario 1 (Fernandez *et al.* 2026, **Figure 2**). The prior for unfished SSB was lognormal(100,000, CV=0.2), process error sigma was set to 0.05, and observation error was estimated from the input CVs plus an estimated variance parameter per series. The demographic analysis with the revised life history scenarios used natural mortality that varies by age and applied to females only. For JABBA-Select it was necessary to have a constant M , so I calculated the values from the assumed maximum ages that would lead to 0.4% of animals surviving to the maximum age to be consistent with the assumptions in the base case. For a sensitivity with lower M , I assumed 1% of sharks would survive to the last age category.

Table 2. Inputs for the JABBA-Select scenarios unchanged from 2025.

Scenario	1	2	3
Linf female	361.527	350	350.3
Linf male	238.49	347.81	241.8
K female	0.103	0.124	0.064
K male	0.24	0.196	0.136
t0 female	-2.29	-1.1	-2.6
t0 male	-1.4	-1	-1.72
aW male	1.25E-05	1.25E-05	1.25E-05
aW female	5.4E-06	5.4E-06	5.4E-06
bW female	3.14	3.14	3.14
bW male	2.97	2.97	2.97
L50_mat	274.1	274.1	274.1
L95_mat	336.3	336.3	336.3

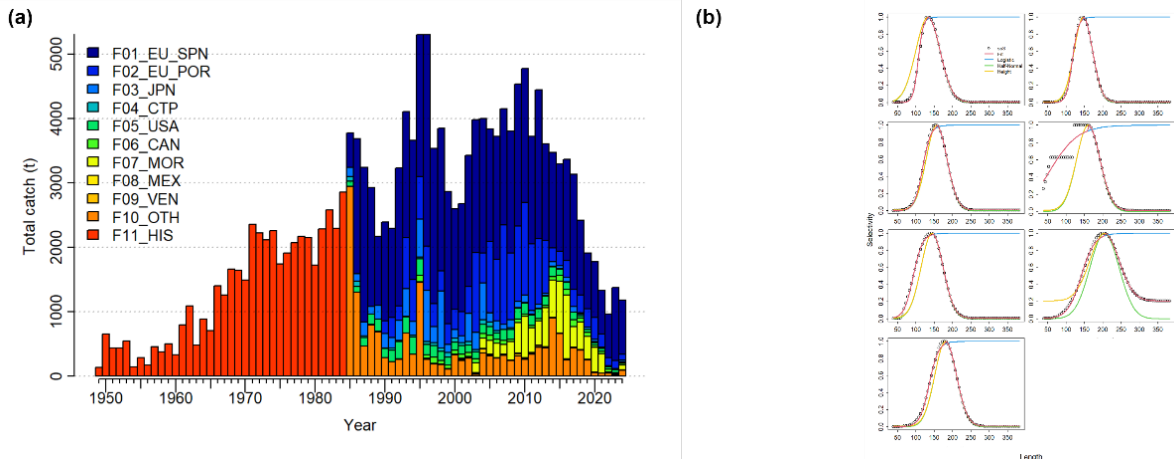


Figure 2. Inputs to JABBA-SELECT, adapted from SS3 Scenario 1 (Fernandez *et al.* 2025) for (a) catches by fleet, and (b) selectivities for EU-Spain, EU-Portugal, Japan, USA, Canada, Mexico and Venezuela.

All analyses were conducted in R with RStudio; GitHub Copilot and Claude AI were used for routine debugging and plotting.

3. Results

3.1 JABBA

The six JABBA runs were consistent in estimating a gradual decline in biomass until the mid-1990s, followed by an increase then decrease so that the current biomass is below the biomass at MSY (B_{msy}) (**Figure 3, Table 3**). However, the scenarios had different productivity. Scenarios base 2, base 3 and sensitivity 3 estimated very low estimated MSY, and very high current harvest rate relative to MSY (H/H_{msy}), while the other scenarios had more plausible H/H_{msy} values. Only scenarios sensitivity 1 and sensitivity 2 estimated $H/H_{msy} < 1$.

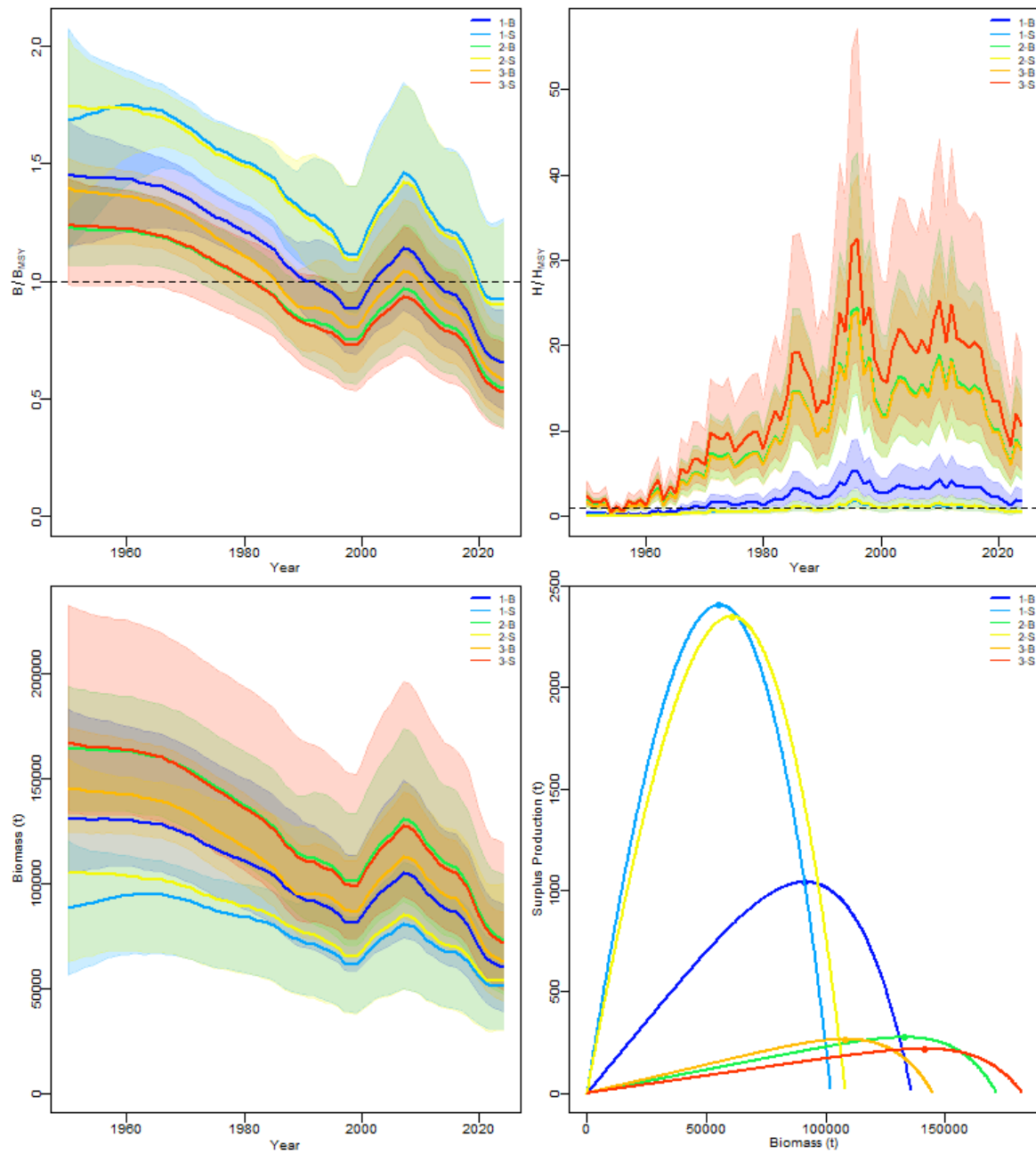


Figure 3. Base (B) and sensitivity (S) JABBA runs (**Table 1**), for updated life history scenarios (1-3).

Table 3. Results from JABBA models for all runs. Current year is 2024.

<i>Scenario</i>	<i>Base</i>				<i>Sensitivity</i>		
	<i>Estimate</i>	<i>mean</i>	<i>lci</i>	<i>uci</i>	<i>mean</i>	<i>lci</i>	<i>uci</i>
1	K	136321	110731	174617	102438	73194	120401
	r	0.06	0.04	0.08	0.11	0.08	0.16
	psi	0.98	0.77	1.13	0.91	0.61	1.12
	sigma.proc	0.05	0.05	0.05	0.05	0.05	0.05
	m	5.04	5.04	5.04	2.48	2.48	2.48
	Hmsy	0.011	0.008	0.016	0.043	0.032	0.063
	SBmsy	91334	74188	116991	55415	39595	65133
	MSY	1054	702	1540	2384	1758	3256
	bmsyk	0.67	0.67	0.67	0.54	0.54	0.54
	P1950	0.97	0.77	1.13	0.91	0.62	1.12
	P2024	0.44	0.31	0.59	0.50	0.36	0.68
	B_Bmsy.cur	0.65	0.46	0.88	0.93	0.66	1.27
	H_Hmsy.cur	1.73	0.97	3.12	0.53	0.31	0.97
2	K	171894	145080	200059	108414	72128	157196
	r	0.02	0.01	0.03	0.10	0.07	0.16
	psi	0.95	0.83	1.11	0.97	0.73	1.14
	sigma.proc	0.05	0.05	0.05	0.05	0.05	0.05
	m	10.00	10.00	10.00	2.69	2.69	2.69
	Hmsy	0.002	0.001	0.003	0.039	0.027	0.059
	SBmsy	133091	112330	154898	60389	40177	87562
	MSY	276	180	424	2336	1670	3511
	bmsyk	0.77	0.77	0.77	0.56	0.56	0.56
	P1950	0.95	0.83	1.11	0.97	0.73	1.13
	P2024	0.42	0.30	0.59	0.50	0.36	0.69
	B_Bmsy.cur	0.54	0.38	0.76	0.90	0.65	1.25
	H_Hmsy.cur	7.87	4.31	14.71	0.56	0.28	0.99
3	K	145166	127094	159602	182691	144496	218531
	r	0.02	0.01	0.03	0.02	0.01	0.02
	psi	1.04	0.86	1.14	0.96	0.76	1.12
	sigma.proc	0.05	0.05	0.05	0.05	0.05	0.05
	m	8.11	8.11	8.11	10.00	10.00	10.00
	Hmsy	0.002	0.002	0.004	0.002	0.001	0.002
	SBmsy	108150	94686	118904	141451	111878	169201
	MSY	263	174	400	216	136	345
	bmsyk	0.75	0.75	0.75	0.77	0.77	0.77
	P1950	1.04	0.86	1.14	0.96	0.76	1.11
	P2024	0.43	0.31	0.61	0.41	0.29	0.57
	B_Bmsy.cur	0.58	0.42	0.81	0.53	0.37	0.74
	H_Hmsy.cur	7.70	4.44	13.66	10.53	5.46	19.17

These differences in productivity were mainly driven by the B_{msy}/K parameter, which was high (>0.7) in scenarios base 2, base 3 and sensitivity 3. In practice, JABBA only allows values of B_{msy}/K up to 0.774 ($m=10$), so scenarios base 2 and sensitivity 3 use 0.774 for B_{msy}/K . The down-up-down pattern is driven by a consistent pattern across the CPUE series, particularly the combined series and Japan 1 (**Figure 4**). However, the fits are not particularly good, in that only Japan 1 and the combined index consistently pass the runs test (**Figure 5**). The posteriors for population growth rate (r) were similar to the priors in all cases (**Figure 6**).

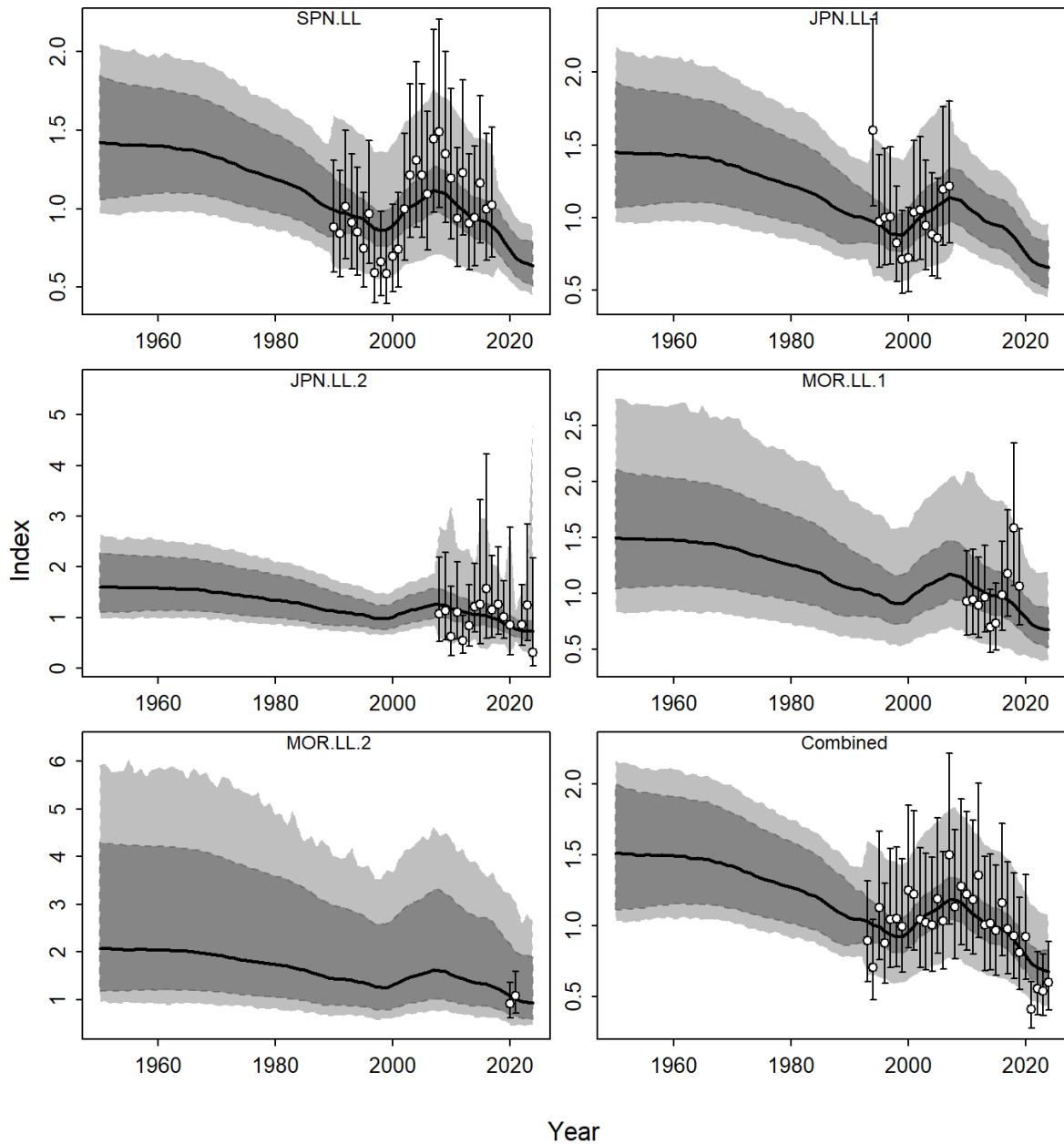


Figure 4. Fits by index for JABBA baseline scenario 1. Other scenarios appear similar.

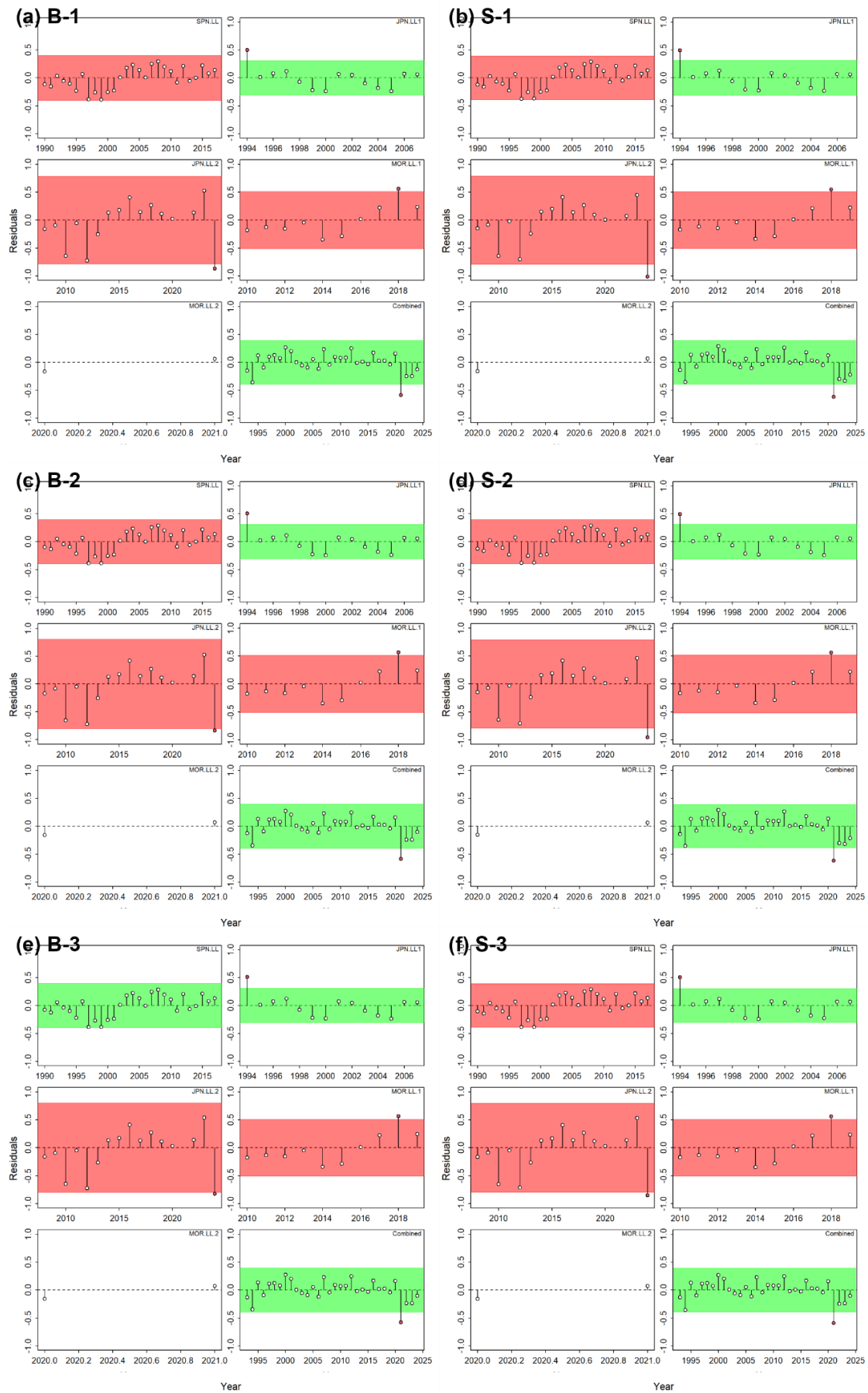


Figure 5. Runs test of base and sensitivity runs for life history scenarios 1 (a), 2 (b) and 3(c) in JABBA.

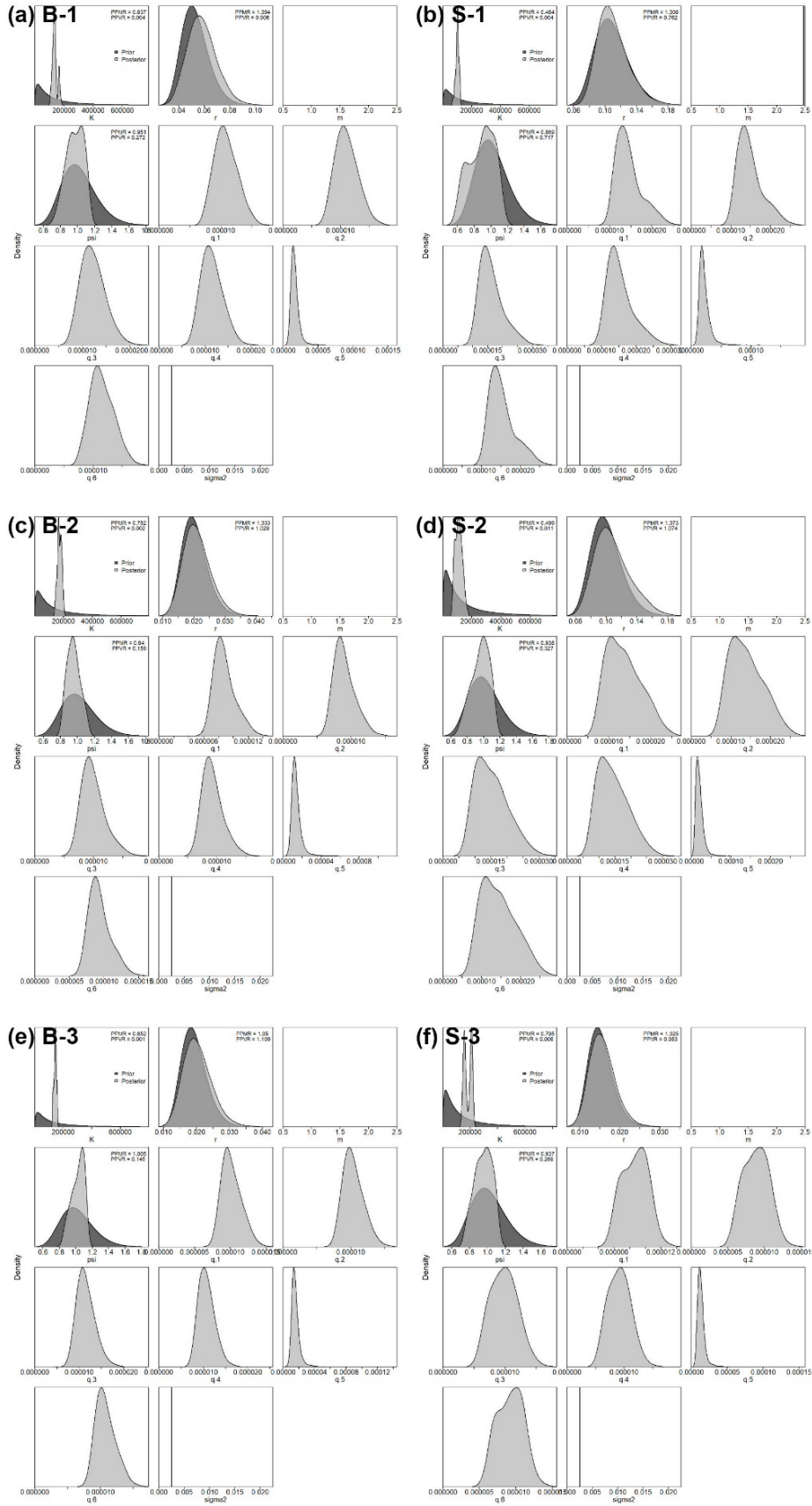


Figure 6. Prior and posterior plots for base and sensitivity runs in JABBA.

The catch only models showed variable productivity of the stock, consistent with the wide range of possible life histories encompassed in the six model configurations (**Figure 7**). Leaving out each series led to fairly consistent trend in biomass except that leaving out the combined index cause the biomass trend to be much more positive in recent years. This is consistent with the fact that the combined index is declining strongly in recent years, while most of the other indices have no values in the recent past because they have been incorporated into the combined index (**Figure 8**).

Retrospective runs ending as far back as 2016 showed roughly consistent relative trends, but with shifts in biomass up and down across the whole time series (**Figure 9, Figure 10, Table 4**). There was also a retrospective bias, with Mohn's Rho values outside the preferred range of $(-0.15 \text{ to } 0.2)$ in several metrics. The prediction skill of the model was poor, especially for the combined index, in which MASE was greater than 1.0 for all scenarios (**Figure 11**).

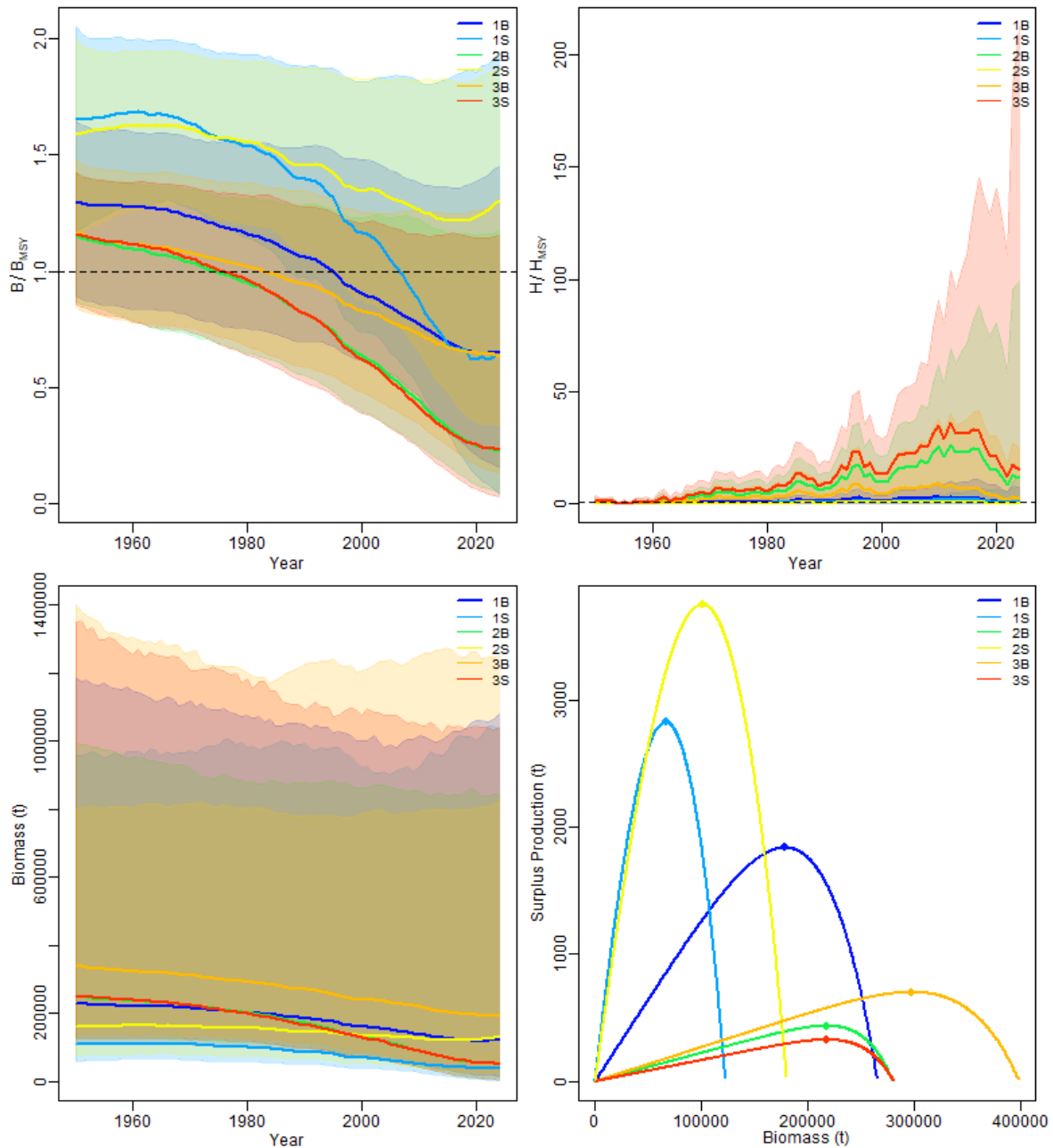


Figure 7. Catch only (post model pre data) runs for all base and sensitivity scenarios in JABBA.

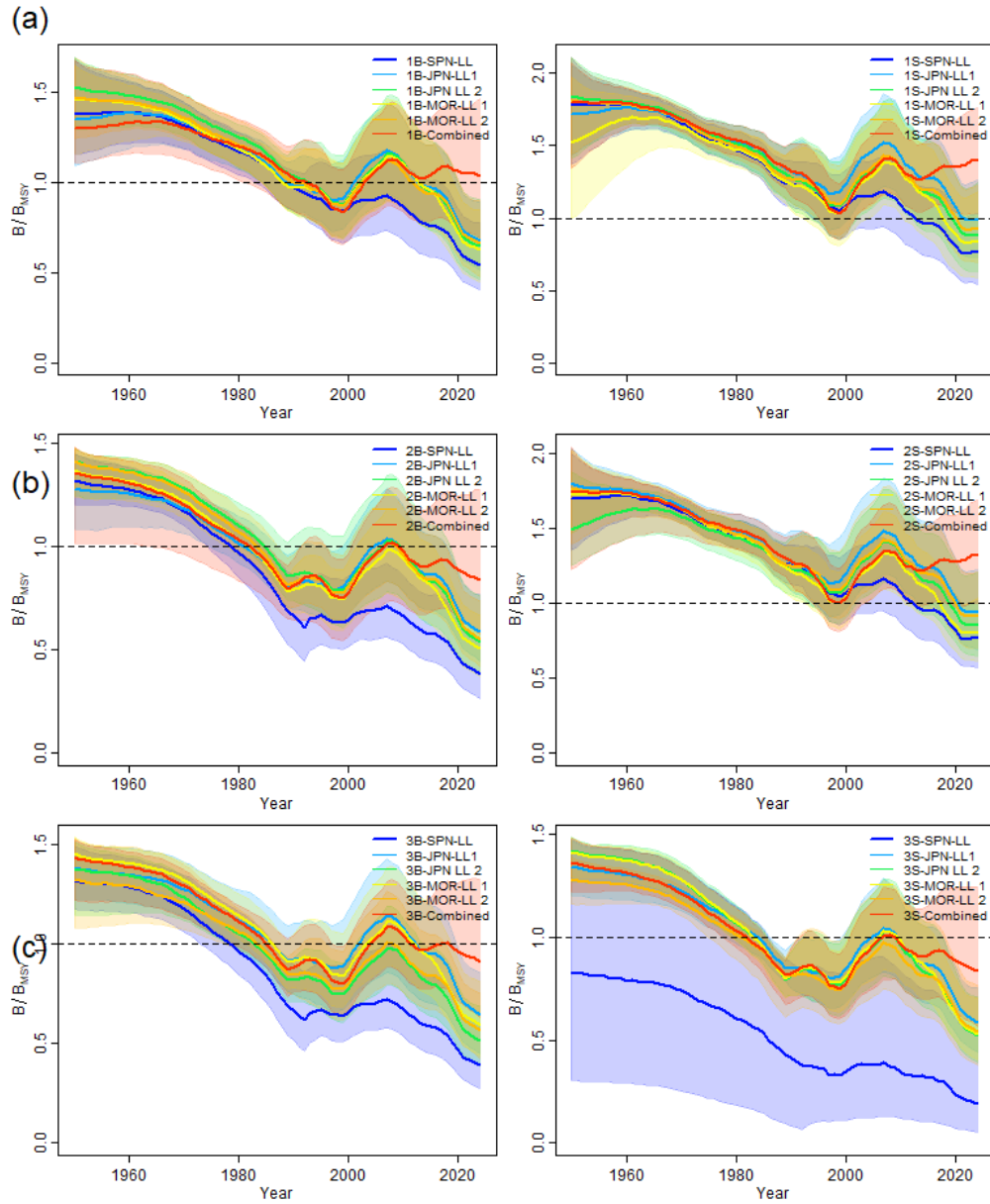
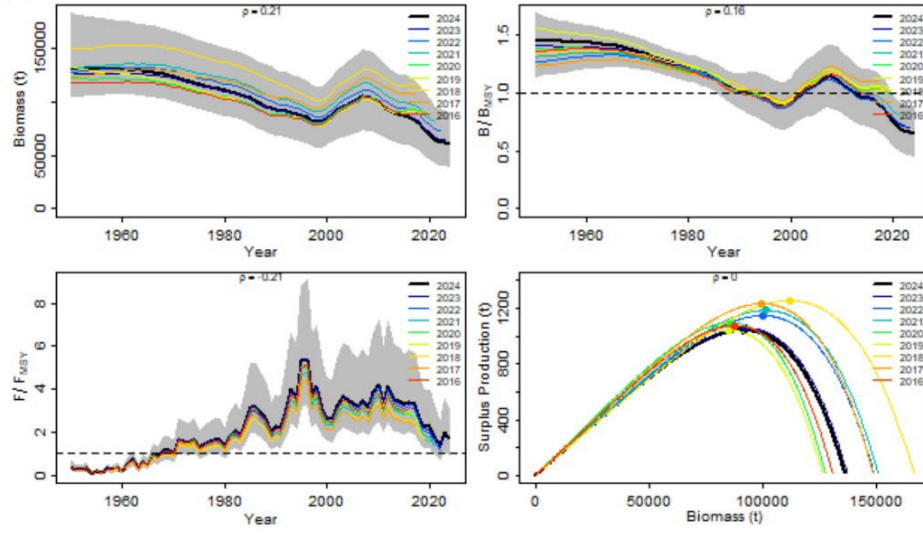


Figure 8. Leave one-out biomass trends for base (B) and sensitivity (S) runs by life history scenario, where the legends indicate the series that was left out.

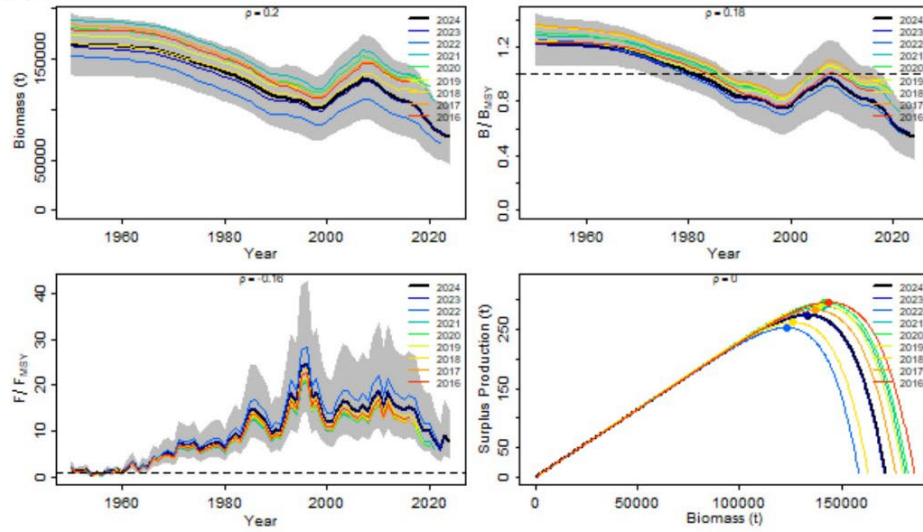
Table 4. Average Mohn's Rho for retrospective analyses in JABBA.

<i>Run</i>	<i>B</i>	<i>F</i>	<i>Bmsy</i>	<i>Fmsy</i>	<i>procB</i>	<i>MSY</i>
Base 1	0.21	-0.17	0.16	-0.21	0.03	0.09
Base 2	0.22	-0.17	0.18	-0.28	0.03	0.19
Base 3	0.20	-0.15	0.18	-0.16	0.02	0.02
Sensitivity 1	0.15	-0.10	0.15	-0.20	0.03	0.07
Sensitivity 2	0.23	-0.18	0.13	-0.20	0.02	0.12
Sensitivity 3	0.23	-0.17	0.15	-0.17	0.02	0.03

(a) B-1



(b) B-2



(c) B-3

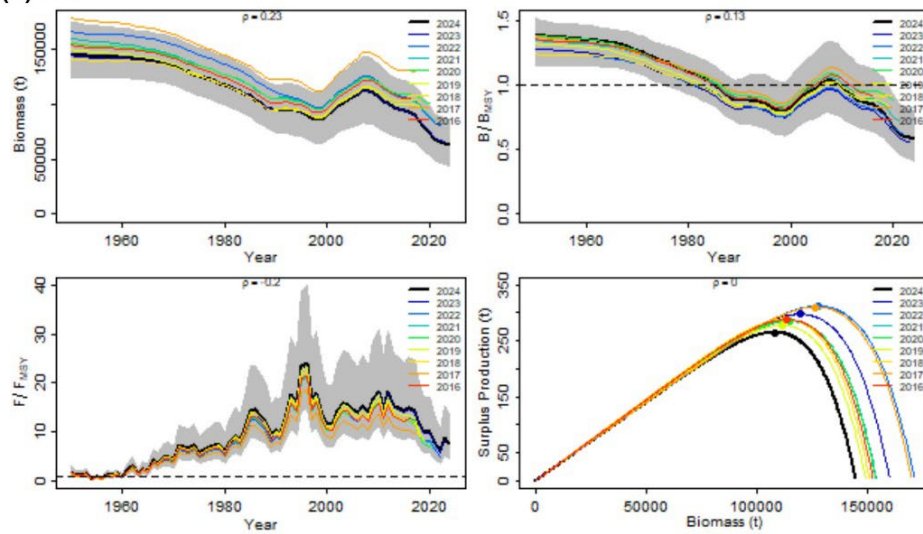
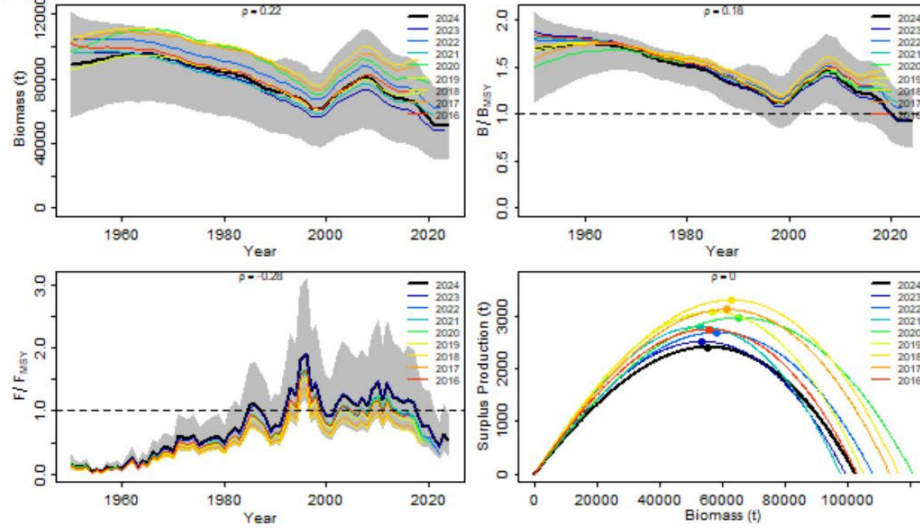
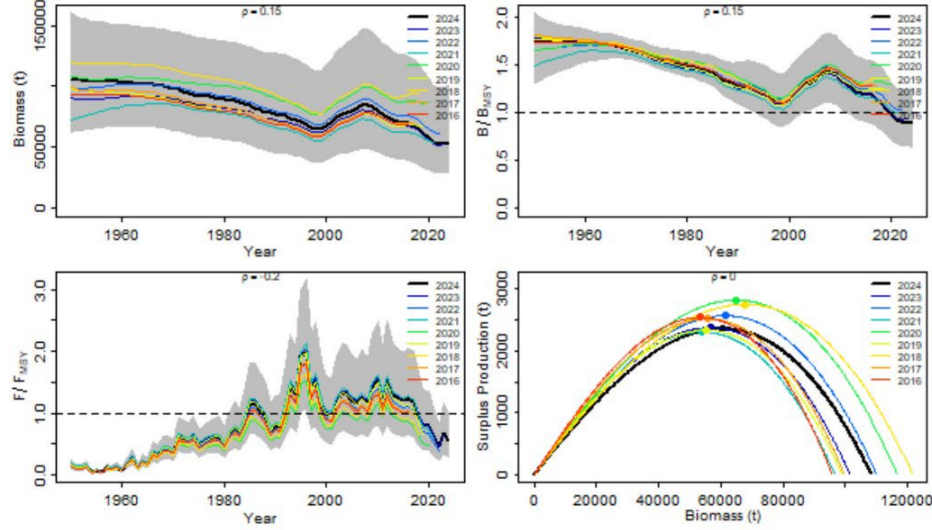


Figure 9. Retrospective plots for base runs for life history scenarios 1 (a), 2 (b) and 3(c) in JABBA.

(a) S-1



(b) S-2



(c) S-3

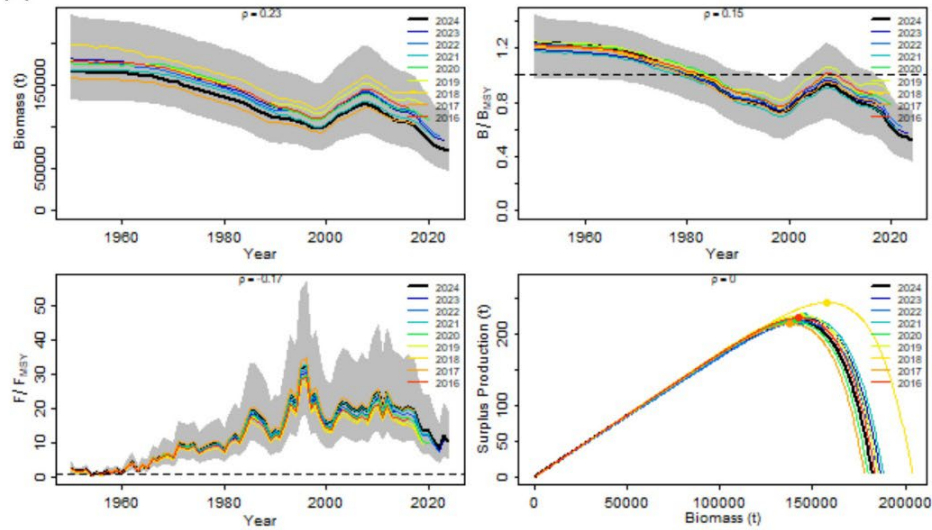


Figure 10. Retrospective plots for sensitivity runs for life history scenarios 1 (a), 2 (b) and 3(c) in JABBA.

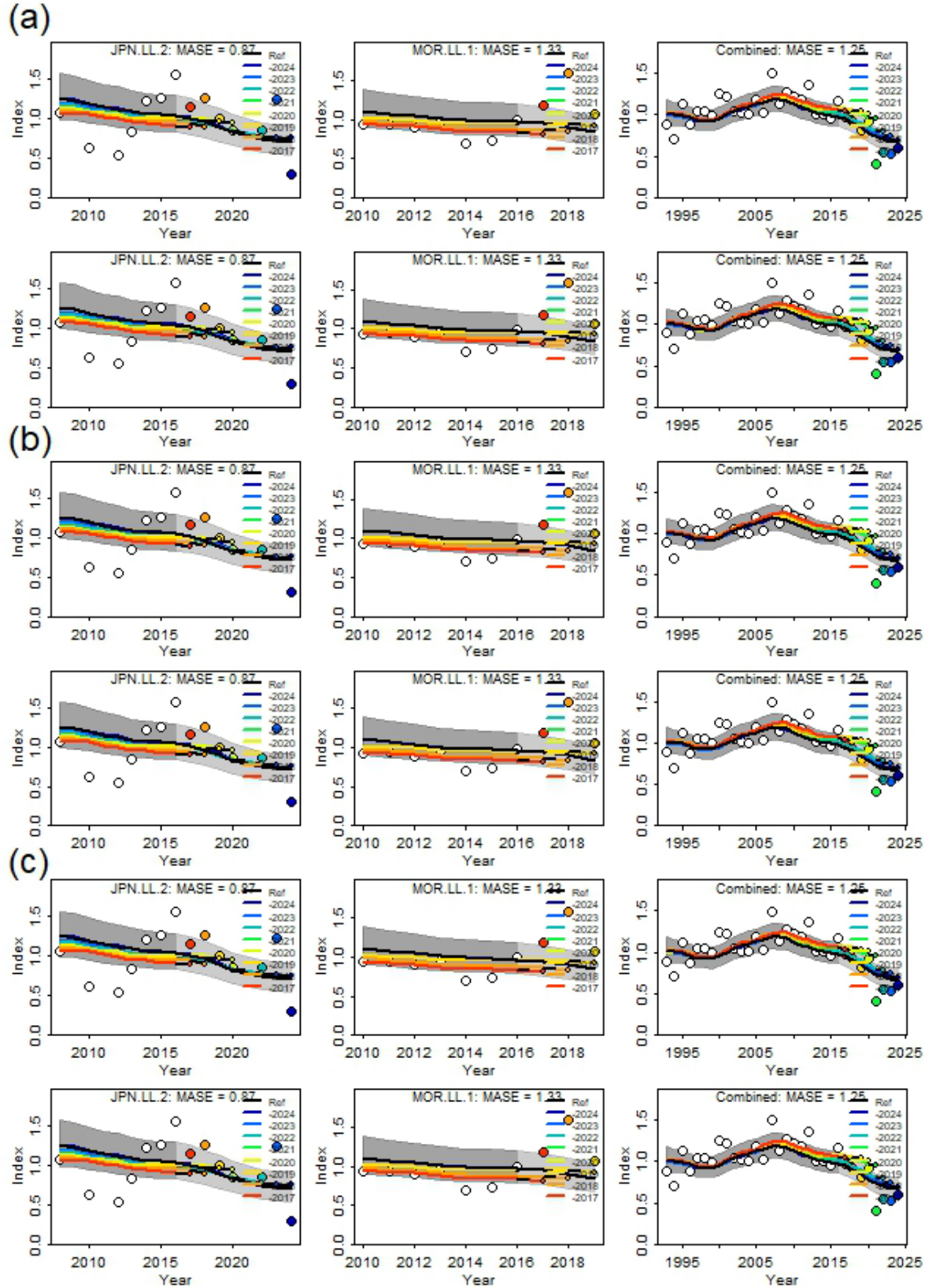


Figure 11. Hindcast cross-validation plots for base and sensitivity runs for life history scenarios 1 (a), 2 (b) and 3(c) in JABBA. In each block, the columns are the three indices that have data in recent years, and the top row is the base, bottom row is sensitivity.

Comparing the results from 2025, the JABBA runs with the 2019 data (Babcock 2025), the current runs, and all the current sensitivities shows that the current scenarios vary widely in productivity, and runs base 2, base 3, and sensitivity 3 have much lower MSY and Hmsy than any previous assessments or sensitivities (**Figure 12**). Thus, only runs base 1, sensitivity 2 and sensitivity 3 are biologically plausible.

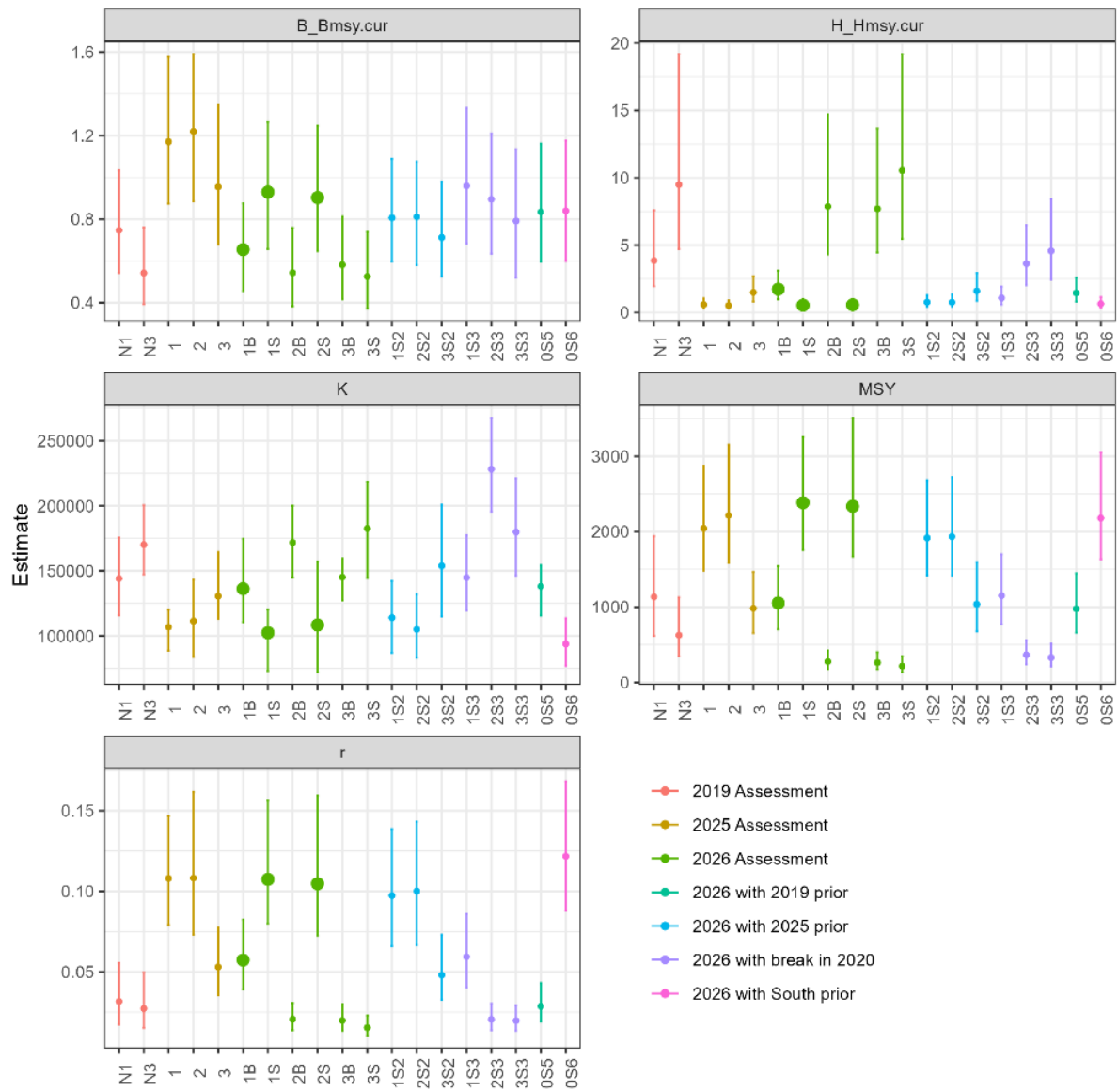


Figure 12. Comparison of current JABBA results with results of the 2019 and 2025 assessments (Babcock, 2025), and the current sensitivities alternative priors.

3.2 JABBA-Select

For JABBA-Select, base scenario 2 could not be run, because the low value of steepness was not consistent with positive surplus production. The other five runs were more plausible, but the yield curves generated by JABBA-Select were different from the values from the demographic analysis, with Bmsy/K around 0.4 to 0.5 for all selectivity patterns (**Figure 13**). As expected, the exploitable biomass was a small fraction of the spawning biomass because the fisheries primarily catch immature sharks.

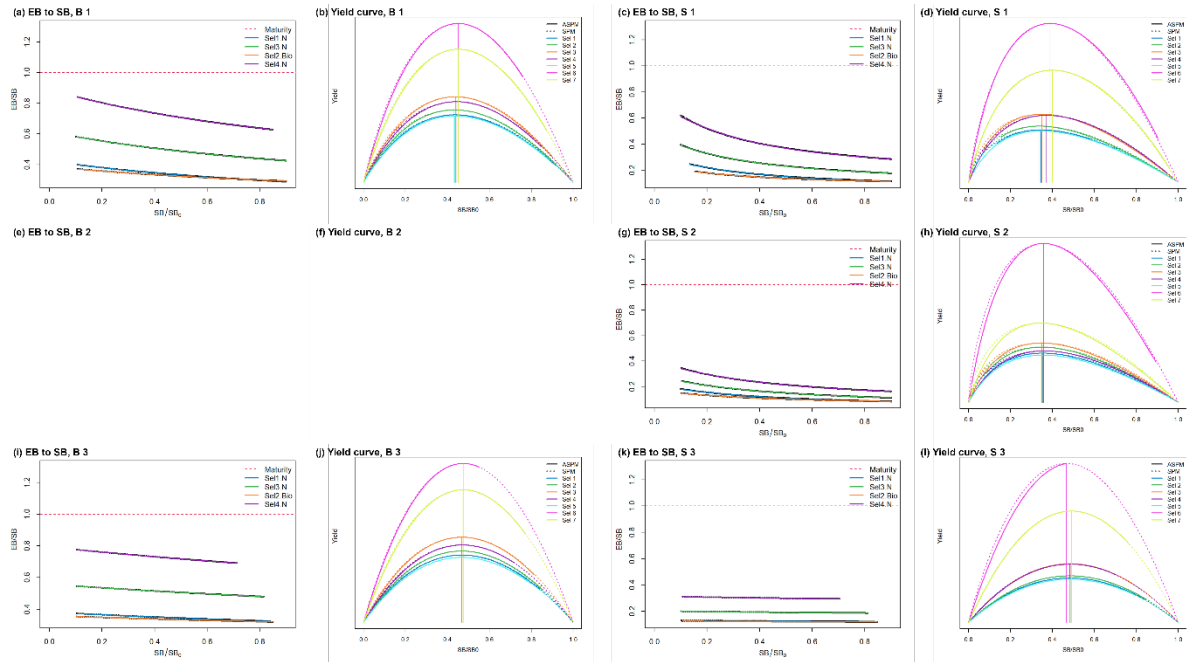


Figure 13. Yield curves and exploitable biomass (EB) to spawning stock biomass (SB) calculated by JABBA-Select for base (B, left) and selectivity (S, right), scenarios 1, 2 and 3. Scenario B2 did not run.

All five runs followed the CPUE indices to generate a down-up-down pattern with current spawning stock biomass close to SSB_{msy} (Figure 14, Figure 15, Table 5). The posteriors show that the K was estimated to be relatively large (Figure 16). The combined series, Japan 1, and Morocco 1 consistently passed the runs test, while the other series occasionally failed (Figure 17).

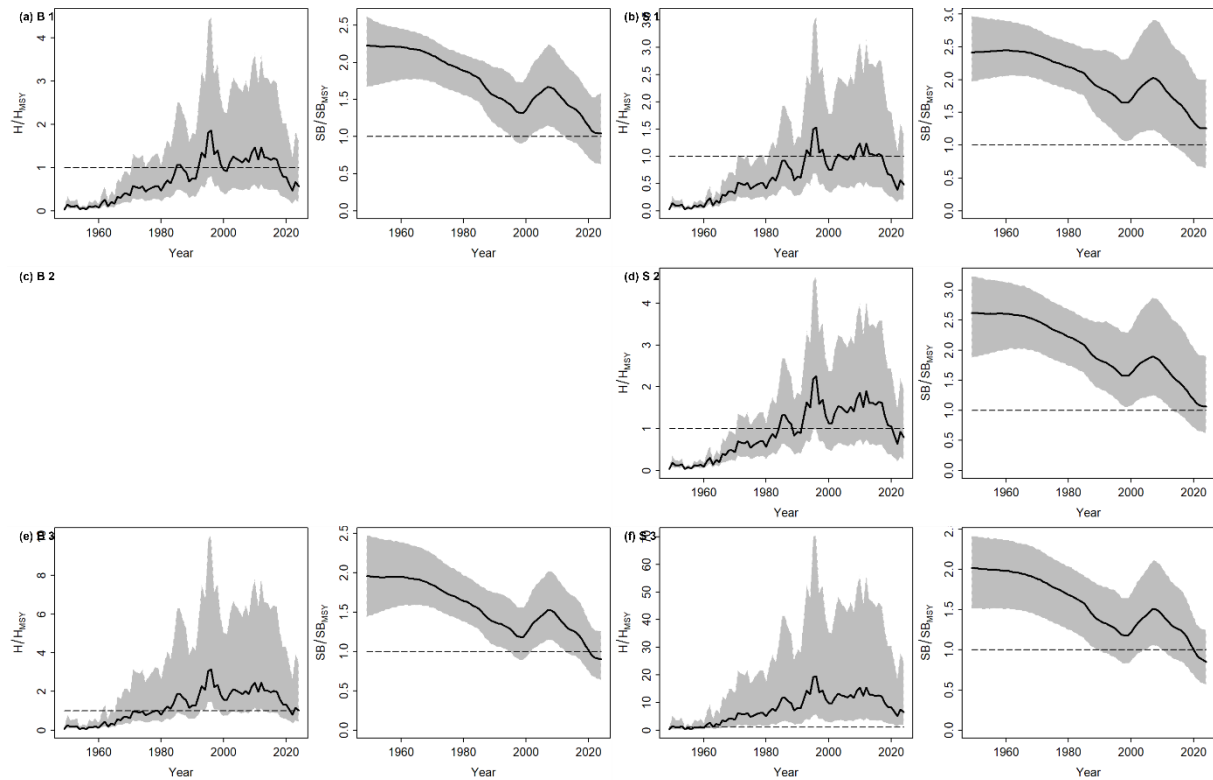


Figure 14. Relative trends for JABBA-Select for base (B, left) and selectivity (S, right), scenarios 1, 2 and 3. Scenario B2 did not run.

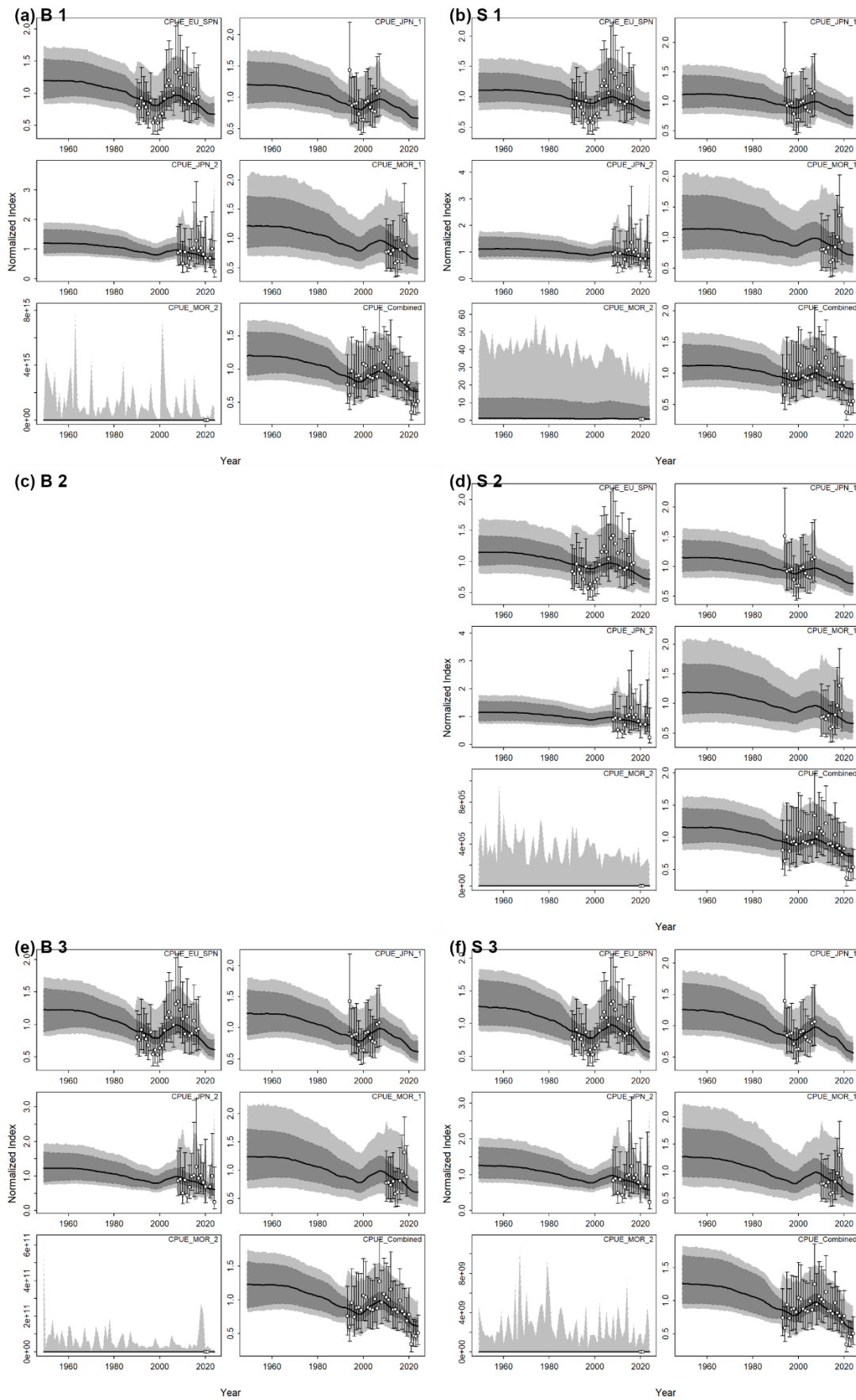


Figure 15. Fits for JABBA-Select for base (B, left) and selectivity (S, right), scenarios 1, 2 and 3. Scenario B2 did not run.

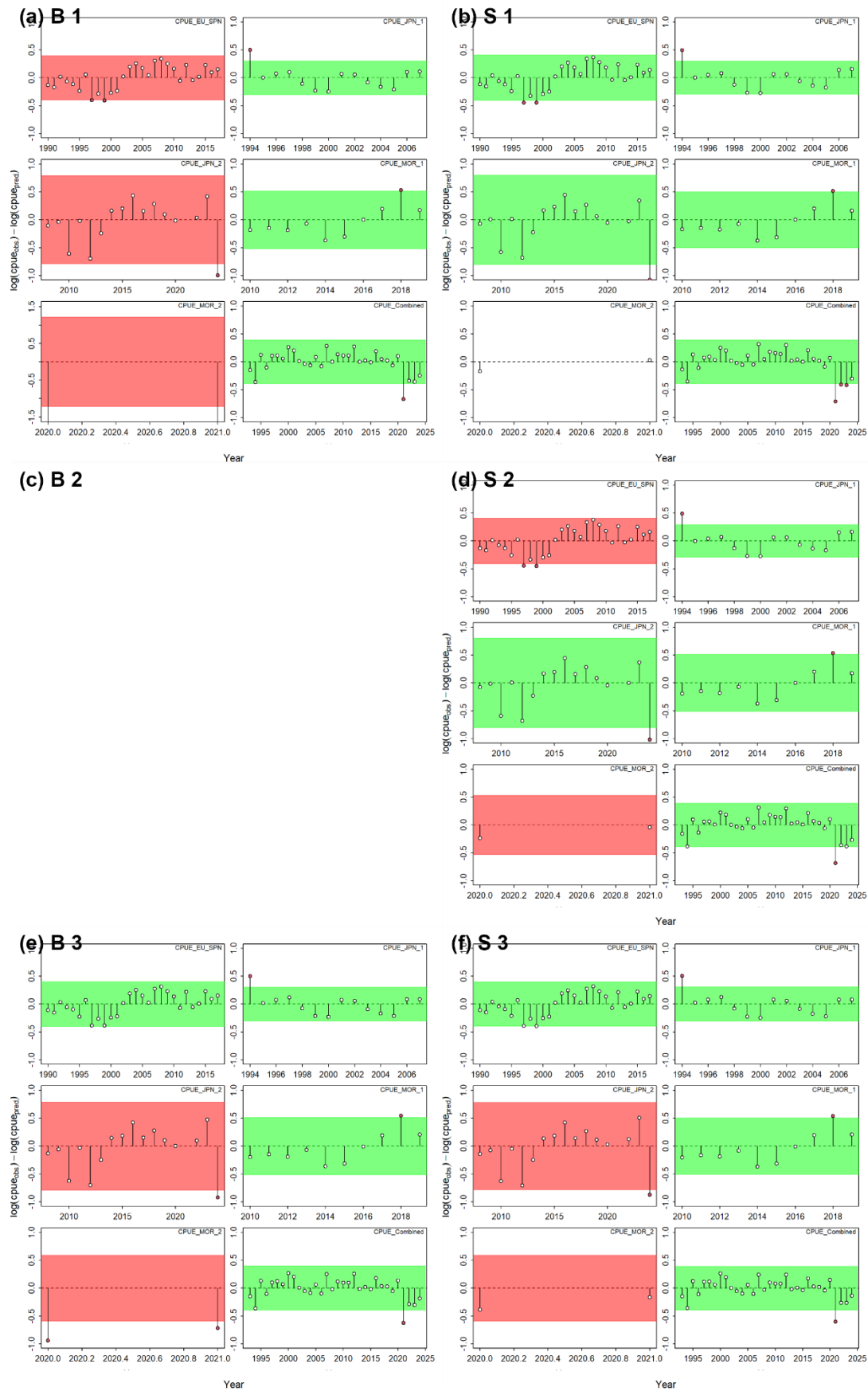


Figure 17. Runs test for JABBA-Select for base (B, left) and selectivity (S, right), scenarios 1, 2 and 3. Scenario B2 did not run.

Table 5. JABBA-select model results. **(a)** Base case

<i>Scenario 1</i>				<i>Scenario 3</i>		
	<i>Median</i>	<i>LCI</i>	<i>UCI</i>	<i>Median</i>	<i>LCI</i>	<i>UCI</i>
SB0	139869	89345	183226	133852	104160	172861
m	1.449	1.252	1.668	1.757	1.204	2.556
Hmsy.1	0.032	0.014	0.089	0.02	0.006	0.045
Hmsy.2	0.035	0.014	0.095	0.022	0.006	0.048
Hmsy.3	0.041	0.017	0.113	0.026	0.007	0.057
Hmsy.4	0.039	0.016	0.108	0.021	0.006	0.054
Hmsy.5	0.032	0.013	0.088	0.019	0.005	0.043
Hmsy.6	0.077	0.03	0.22	0.037	0.009	0.118
Hmsy.7	0.064	0.027	0.18	0.037	0.01	0.096
SBmsy	61715	37663	82055	62897	49112	83291
MSY.1	1983	935	3839	1309	393	2581
MSY.2	2126	1002	4108	1389	417	2748
MSY.3	2510	1183	4853	1635	490	3256
MSY.4	2377	1108	4649	1337	371	3184
MSY.5	1948	921	3768	1214	360	2503
MSY.6	4675	2092	9664	2363	557	6975
MSY.7	3933	1804	7772	2375	658	5551
Href.1	0.032	0.014	0.089	0.02	0.006	0.045
Href.2	0.035	0.014	0.095	0.022	0.006	0.048
Href.3	0.041	0.017	0.113	0.026	0.007	0.057
Href.4	0.039	0.016	0.108	0.021	0.006	0.054
Href.5	0.032	0.013	0.088	0.019	0.005	0.043
Href.6	0.077	0.03	0.22	0.037	0.009	0.118
Href.7	0.064	0.027	0.18	0.037	0.01	0.096
SBref	61715	37663	82055	62897	49112	83291
Yref.1	1983	935	3839	1309	393	2581
Yref.2	2126	1002	4108	1389	417	2748
Yref.3	2510	1183	4853	1635	490	3256
Yref.4	2377	1108	4649	1337	371	3184
Yref.5	1948	921	3768	1214	360	2503
Yref.6	4675	2092	9664	2363	557	6975
Yref.7	3933	1804	7772	2375	658	5551
P1949	0.973	0.739	1.122	0.928	0.711	1.101
P2024	0.459	0.276	0.674	0.424	0.309	0.593
B_Bmsy.cur	1.044	0.623	1.597	0.899	0.633	1.259
H_Hmsy.cur	0.572	0.203	1.615	1.012	0.414	3.482

Table 5, continued (b) Sensitivity

<i>Scenario 1</i>				<i>Scenario 2</i>			<i>Scenario 3</i>		
	<i>Median</i>	<i>LCI</i>	<i>UCI</i>	<i>Median</i>	<i>LCI</i>	<i>UCI</i>	<i>Median</i>	<i>LCI</i>	<i>UCI</i>
SB0	190066	175809	197521	160647	127407	184917	207517	169747	235289
m	1	0.811	1.225	0.912	0.736	1.128	1.703	1.419	2.052
Hmsy.1	0.028	0.014	0.054	0.025	0.012	0.05	0.002	0.001	0.008
Hmsy.2	0.03	0.015	0.058	0.028	0.014	0.056	0.002	0.001	0.008
Hmsy.3	0.036	0.018	0.07	0.031	0.015	0.06	0.003	0.001	0.011
Hmsy.4	0.036	0.018	0.069	0.027	0.013	0.052	0.003	0.001	0.01
Hmsy.5	0.027	0.014	0.053	0.024	0.012	0.048	0.002	0.001	0.008
Hmsy.6	0.084	0.041	0.167	0.082	0.039	0.165	0.004	0	0.026
Hmsy.7	0.059	0.03	0.116	0.041	0.02	0.08	0.005	0.002	0.02
SBmsy	69138	60357	77916	56302	42589	68211	96908	80067	111334
MSY.1	1917	1059	3467	1398	742	2527	209	56	777
MSY.2	2052	1131	3714	1576	840	2843	219	59	815
MSY.3	2485	1372	4503	1689	896	3049	276	74	1024
MSY.4	2459	1344	4465	1468	783	2670	255	67	958
MSY.5	1877	1036	3400	1346	715	2441	203	54	755
MSY.6	5827	3093	10816	4517	2365	8343	413	47	2542
MSY.7	4103	2245	7472	2247	1197	4061	522	137	1920
Href.1	0.028	0.014	0.054	0.025	0.012	0.05	0.002	0.001	0.008
Href.2	0.03	0.015	0.058	0.028	0.014	0.056	0.002	0.001	0.008
Href.3	0.036	0.018	0.07	0.031	0.015	0.06	0.003	0.001	0.011
Href.4	0.036	0.018	0.069	0.027	0.013	0.052	0.003	0.001	0.01
Href.5	0.027	0.014	0.053	0.024	0.012	0.048	0.002	0.001	0.008
Href.6	0.084	0.041	0.167	0.082	0.039	0.165	0.004	0	0.026
Href.7	0.059	0.03	0.116	0.041	0.02	0.08	0.005	0.002	0.02
SBref	69138	60357	77916	56302	42589	68211	96908	80067	111334
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Yref.3	2485	1372	4503	1689	896	3049	276	74	1024
Yref.4	2459	1344	4465	1468	783	2670	255	67	958
Yref.5	1877	1036	3400	1346	715	2441	203	54	755
Yref.6	5827	3093	10816	4517	2365	8343	413	47	2542
Yref.7	4103	2245	7472	2247	1197	4061	522	137	1920
P1949	0.886	0.737	1.06	0.926	0.658	1.102	0.946	0.731	1.101
P2024	0.461	0.253	0.713	0.374	0.221	0.659	0.399	0.268	0.58
B_Bmsy.cur	1.256	0.654	2.003	1.063	0.622	1.913	0.851	0.561	1.248
H_Hmsy.cur	0.481	0.188	1.406	0.797	0.268	1.939	6.543	1.745	24.835

Comparing the results from 2026 to those from 2025 shows that the current runs are more depleted and have higher fishing mortality relative to MSY level, despite having relatively higher unfished SSB (**Figure 18**). The values of MSY, which vary by fleet depending on selectivity were more variable in the current runs than in 2025.

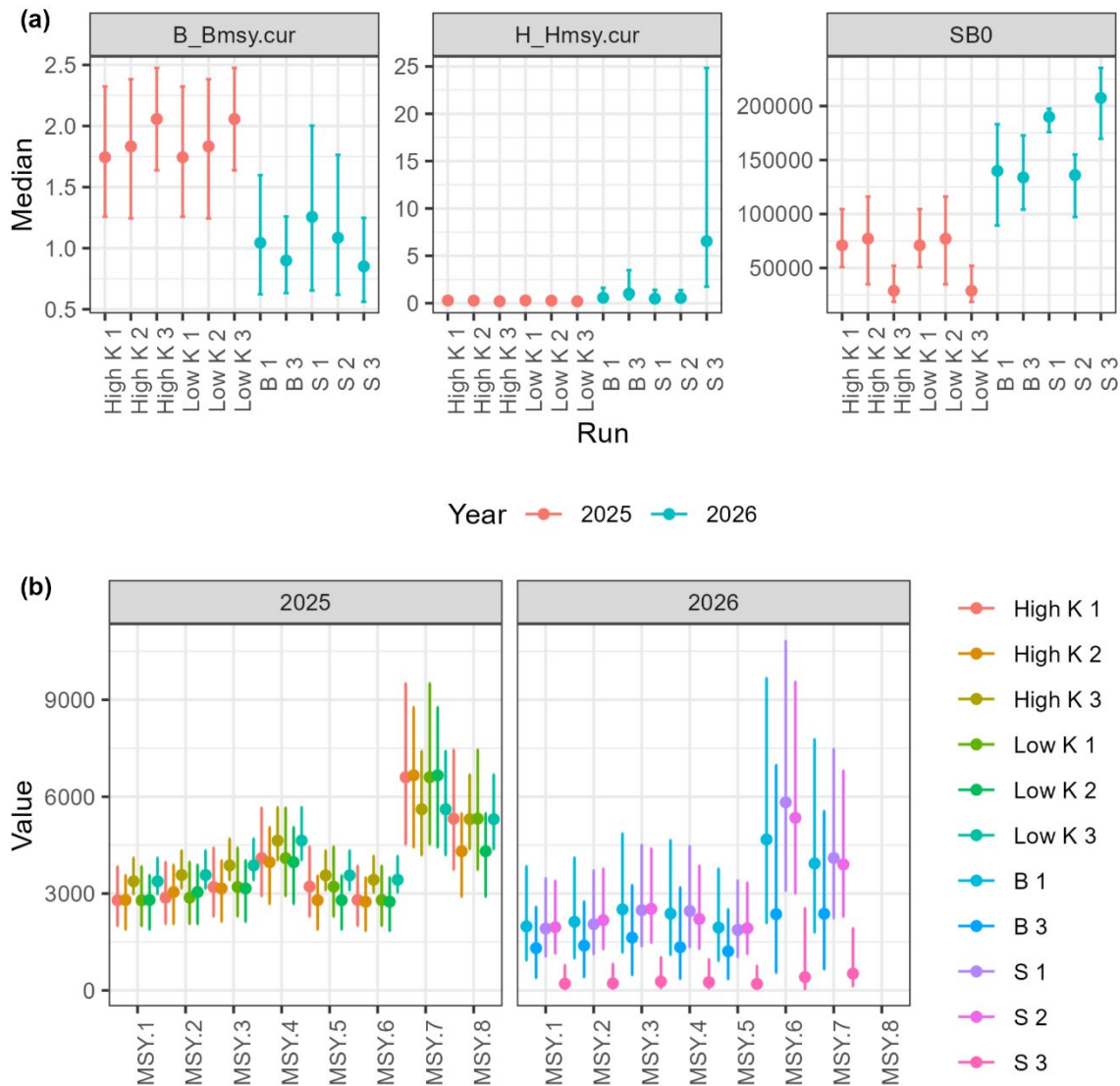


Figure 18. Comparing 2025 and 2026 JABBA-Select runs (a) across scenarios for status and unfished SSB, and (b) comparing the MSY estimates across fleets. The fleets are not the same in 2025 and 2026.

4. Discussion

For both JABBA and JABBA-Select the results differ from the 2025 assessment, because of different and more consistent trends in the CPUE series, and very different life history assumptions. Across all the JABBA and JABBA-Select runs, the trends in biomass were consistent, and implied a decline in recent years provided the combined index was included in the data. However, the values of, r MSY and the unfished biomass (K in JABBA, $SB0$ in JABBA-Select) were highly variable across life history scenarios because the priors were so different, leading to different conclusions about overfishing status. Thus, the uncertainty about the life history of the north Atlantic shortfin mako population is the largest source of uncertainty in the production model assessments.

For both models, because the fisheries capture almost no mature individuals, the estimated status, trends and reference points should be considered highly uncertain because of the lack of data on mature animals. Also, the production models do not adequately capture the time lag between fishing mortality and the potential impact on spawning stock biomass for a stock where only juveniles are caught. Thus, these models should not be used for projections into the future. Nevertheless, they are able to capture trends in exploitable biomass and demonstrate the stock is apparently in a downward trajectory in recent years.

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